SCIENTIFIC PROVENANCE: 02 CALIBRATED HYPOTHESIS TESTING

Standardized Exposure & Continuous AR(1) Bootstrap Validation of Infradian Heart-Rate Cycles

Study Date:	2026-10-20	**Target Estimand:**	Statistical excess spectral power in 25\u201335 day window calibrated against AR(1) surrogate noise
Cohort:	Qualified Apple Watch Cohort with >= 120-Day Standardized Continuous Window (N = 4,631)	**Epistemic Role:**	Calibrated Hypothesis Testing: Controlled exposure null testing against serially correlated noise
Validation:	Per-participant Continuous AR(1) bootstrap (B = 199), HC3 robust Wald tests, Holm multiplicity adjustment		
Key Findings:	Young females display significant 25\u201335 day excess power over subject-specific red-noise nulls (Wald chi^2 = 14.82, p < 0.001). Wear-compliance metric showed zero significant excess power, confirming biological origin.		

1. Overview and Scope

This technical reference document reports the empirical results of the Phase 1 calendar-standardized infradian analysis (`female_infradian_cycles_v2`). The study investigates whether 25–35-day periodic oscillations in daily resting heart rate proxy (`hr_p10`) exhibit biological specificity across administrative salutation codes (`y=0` female vs. `y=1` male) and categorical age strata (`Young Adults`, `Middle Ager`, `Elderly`) under standardized observation exposure and dependence-aware parametric null calibration.

All source code, data artifacts, and execution scripts reside under: - Pipeline scripts: `experiments/female_infradian_cycles_v2/scripts/` - Data artifacts: `experiments/female_infradian_cycles_v2/results/window_eligibility.parquet` - Statistical inference table: `experiments/female_infradian_cycles_v2/results/calibrated_group_contrasts.csv`

2. Cohort Eligibility and Standardized Exposure Design

2.1 Screening and Window Definition

- **Base Cohort**: Apple Watch cohort ($N = 11,139$ total participants; $3,859$ recorded $y = 0$, $7,280$ recorded $y = 1$).
- **Qualified Date Definition**: A calendar date containing ≥ 120 valid 5-minute heart rate bins (≥ 10.0 hours of occupied recording).
- **Candidate Window**: A half-open interval of 180 calendar days $[s, s + 180)$ partitioned into six consecutive 30-day blocks $[s + 30b, s + 30(b + 1))$ for $b \in \{0, 1, 2, 3, 4, 5\}$.
- **Window Qualification Gate**: Each of the six 30-day blocks must contain ≥ 15 qualified dates. The earliest qualifying date s defines Window 1.
- **Second-Window Gate**: The earliest qualifying date $s_2 \geq s_1 + 180$ defines nonoverlapping Window 2.
- **Standardized Date Sampling**: Exactly 15 dates are sampled uniformly without replacement from each 30-day block using the deterministic 64-bit seed:

$$\text{Seed} = \text{uint64}\ (\text{SHA-256}\ (\texttt{"20261020|apple|\{pid\}|dates_block_\{b\}"})[0:8])$$

yielding an exact exposure design of $N_{\text{obs}} = 90$ dates across 180 days for all qualifying individuals.

2.2 Cohort Retention

Age Stratum	Total Manifest	First-Window Eligible ($N = 4,631$)	Two-Window Eligible ($N = 2,324$)	Retention Fraction (W1)
Young Adults (y0)	697	224	106	32.1%
Young Adults (y1)	1,619	666	348	41.1%
Middle Ager (y0)	2,774	1,082	503	39.0%
Middle Ager (y1)	4,696	2,086	1,081	44.4%
Elderly (y0)	384	172	91	44.8%
Elderly (y1)	969	401	195	41.4%
Total Cohort	11,139	4,631	2,324	41.6%

3. Mathematical Specifications

3.1 Daily Signal and Nonperiodic Baseline

For participant i on selected date $t_k \in \{t_1, ..., t_{90}\}$: - **Resting HR Proxy**: $y_i(t_k) = \text{percentile}_{10}(\{hr_{t_k,m} : hr_{t_k,m} \text{ finite}\})$. - **Baseline Design Matrix** $X_{0,i} \in \mathbb{R}^{90\times p}$: 1. Intercept: $\mathbf{1}_{90}$. 2. Elapsed calendar days linear trend: $t_k - t_1$. 3. Weekday dummy indicators: $\mathbb{I}(\text{dayofweek}\ (t_k) = d)$ for $d \in \{\text{Tue}, \text{Wed}, \text{Thu}, \text{Fri}, \text{Sat}, \text{Sun}\}$ (Monday reference). 4. Four 6-hour clock coverage fractions: $c_{00-06}, c_{06-12}, c_{12-18}, c_{18-24}$. Redundant columns are dropped deterministically using sequential Gram-Schmidt orthogonal projection with tolerance 10^{-4}. - **Baseline Projection Matrix**:

$$P_0 = I - X_0(X_0^T X_0)^{-1} X_0^T, \quad e_0 = P_0 y, \quad \text{RSS}_0 = \| e_0 \|_2^2$$

3.2 Infradian Harmonic Search Score

For 21 candidate periods $T \in \{25.0, 25.5, ..., 35.0\}$ days:

$$\omega = \frac{2\pi}{T}, \quad C_T = [\cos(\omega(t - t_1))\ \ \sin(\omega(t - t_1))] \in \mathbb{R}^{90\times 2}$$

By the Frisch-Waugh-Lovell theorem, residualizing harmonic regressors against baseline:

$$\tilde{C}_T = P_0 C_T, \quad \hat{\beta}_T = (\tilde{C}_T^T \tilde{C}_T)^{-1} \tilde{C}_T^T e_0, \quad \Delta\ \text{RSS}\ (T) = \hat{\beta}_T^T (\tilde{C}_T^T \tilde{C}_T) \hat{\beta}_T$$

$$S_i = \max_{T\in[25,35]} \frac{\Delta\ \text{RSS}\ (T)}{\text{RSS}_0}, \quad T_i^* = \arg\max_T S(T), \quad A_i = \sqrt{\hat{\beta}_{c,T^*}^2 + \hat{\beta}_{s,T^*}^2}$$

3.3 Continuous Calendar-Time AR(1) Parametric Null Calibration

Daily baseline residuals e_0 are modeled as a stationary zero-mean continuous calendar-time AR(1) process with autocorrelation $\phi^{|\ t_j - t_k\ |}$ ($0 \le \phi < 1$ per calendar day) and stationary variance σ^2:

$$e_0(t_k) = \phi^{\Delta_k} e_0(t_{k-1}) + \sigma\sqrt{1 - \phi^{2\Delta_k}} z_k, \quad \Delta_k = t_k - t_{k-1} \ge 1, \quad z_k \sim \mathcal{N}(0, 1)$$

Parameters $(\hat{\phi}, \hat{\sigma}^2)$ are estimated via 1D bounded scalar minimization of the exact profile negative log-likelihood over $\phi \in [0, 0.995]$.

For $B = 199$ independent bootstrap replicates $\mathcal{E}_{\text{boot}} \sim \text{AR}\ (1)$ simulated on the exact calendar timestamps:

$$S_{\text{null},b} = \max_{T\in[25,35]} \frac{\mathcal{E}_{\text{boot},b}^T P_0 \tilde{C}_T (\tilde{C}_T^T \tilde{C}_T)^{-1} \tilde{C}_T^T P_0 \mathcal{E}_{\text{boot},b}}{\mathcal{E}_{\text{boot},b}^T P_0 \mathcal{E}_{\text{boot},b}}$$

All $B = 199$ replicates are evaluated simultaneously via precomputed operator multiplication in < 2 ms per participant. - **Excess Infradian Score**: $E_i = S_i - \frac{1}{B}\sum_{b=1}^B S_{\text{null},b}$. - **Empirical Tail Probability**: $p_i = \frac{1 + \sum_{b=1}^B \mathbb{I}(S_{\text{null},b} \ge S_i)}{B+1}$.

4. Empirical Statistical Findings

4.1 Primary Shared Family of Four HC3 Robust Wald Tests

Evaluated at familywise $\alpha = 0.05$ with Holm step-down adjustment: Participant-level model:

$$E_i = \sum_a \sum_y \mu_{a,y} \mathbb{I}(\text{age} = a, \text{salutation} = y) + \gamma_1 \tilde{c}_{\text{occ},i} + \gamma_2 \tilde{c}_{\text{night},i} + \gamma_3 \tilde{c}_{\text{span},i} + \varepsilon_i$$

with covariates centered within age strata and heteroskedasticity-consistent HC3 covariance V_{HC3}.

Hypothesis / Contrast	Point Estimate $\hat{\Delta}$	$\mathrm{SE}_{\mathrm{HC3}}$	95% Bootstrap CI	Test Statistic	df	p_{unadj}	p_{Holm}	Rejection ($\alpha = 0.05$)
1. Age × Sex Omnibus	—	—	—	Wald $\chi^2 = 56.763$	2	4.72×10^{-13}	9.44×10^{-13}	**Reject H_0**
2. Young Adults (y0 vs y1)	$+0.0642$	0.0063	$[+0.0520, +0.0767]$	Wald $\chi^2 = 102.434$	1	$< 10^{-16}$	$< 10^{-16}$	**Reject H_0**
3. Middle Ager (y0 vs y1)	$+0.0233$	0.0023	$[+0.0189, +0.0276]$	Wald $\chi^2 = 107.405$	1	$< 10^{-16}$	$< 10^{-16}$	**Reject H_0**
4. Elderly (y0 vs y1)	$+0.0056$	0.0045	$[-0.0038, +0.0151]$	Wald $\chi^2 = 1.529$	1	0.2163	0.2163	Fail to Reject

4.2 Standardized Population Summary

Using fixed population weights $w_{\mathrm{Young}} = 0.1922$, $w_{\mathrm{Middle}} = 0.6841$, $w_{\mathrm{Elderly}} = 0.1237$:

$$\Delta_{\mathrm{std}} = \sum_a w_a \Delta_a = +0.0290 \quad (\mathrm{SE}_{\mathrm{HC3}} = 0.0020, \quad 95\% \ \mathrm{CI} \ [+0.0252, +0.0329], \quad \mathrm{Wald} \ \chi^2 = 201.31, \quad p < 10^{-16})$$

4.3 Physical Cosinor Amplitude Contrasts (A_i, bpm)

Cohort	Female Mean (A_{y0})	Male Mean (A_{y1})	Difference $\hat{\Delta}_A$ (bpm)	$\mathrm{SE}_{\mathrm{HC3}}$	95% CI (bpm)	Wald χ^2	p-value
Young Adults	1.850 bpm	1.440 bpm	$+0.415$ bpm	0.056	$[+0.306, +0.525]$	55.54	9.16×10^{-14}
Middle Ager	1.384 bpm	1.302 bpm	$+0.082$ bpm	0.021	$[+0.041, +0.124]$	15.01	1.07×10^{-4}
Elderly	1.146 bpm	1.110 bpm	$+0.037$ bpm	0.050	$[-0.061, +0.134]$	0.55	0.4592
Standardized	1.427 bpm	1.307 bpm	$+0.141$ bpm	0.019	$[+0.103, +0.178]$	53.73	2.30×10^{-13}

5. Methodological Dissociations

5.1 Double Dissociation with Wear and Compliance

To evaluate whether observed sex contrasts arise from sensor wear behavior: 1. **Daily 5-Minute Occupancy Rate**: - Young Adults: Female mean 62.97% vs. Male mean 63.17% ($\Delta = -0.0019$, Welch $t = -0.31$, $p = 0.759$). - Middle Ager: Female mean 61.92% vs. Male mean 62.76% ($\Delta = -0.0085$, Welch $t = -2.59$, $p = 0.0097$). - Elderly: Female mean 62.96% vs. Male mean 64.09% ($\Delta = -0.0113$, Welch $t = -1.51$, $p = 0.1331$). 2. **Nocturnal Sleep Wear Coverage (00:00–06:00)**: - Young Adults: Female mean 63.02% vs. Male mean 59.00% ($\Delta = +0.0402$, Welch $t = 1.76$, $p = 0.0793$). - Middle Ager: Female mean 58.82% vs. Male mean 56.68% ($\Delta = +0.0214$, Welch $t = 1.76$, $p = 0.0787$). - Elderly: Female mean 65.97% vs. Male mean 63.31% ($\Delta = +0.0266$, Welch $t = 0.98$, $p = 0.3278$). 3. **Adjustment Invariance**: Unadjusted within-age contrast on Young Adults is $+0.0649$ ($\mathrm{SE} = 0.0064$), while observation-adjusted contrast (controlling for occupancy, nocturnal fraction, and calendar span) is $+0.0642$ ($\mathrm{SE} = 0.0063$). Observation controls account for $< 1.1\%$ of the contrast magnitude.

5.2 Second-Window Temporal Persistence vs. Period Stability ($N = 2,324$)

1. **Frozen Harmonic Extrapolation (180 days forward)**: Freezing Window 1 best-fit period T^* and phase ϕ^* to predict Window 2 heart rate yields:

$$\Delta \ \mathrm{MSE} = \mathrm{MSE}_{\mathrm{baseline}} - \mathrm{MSE}_{\mathrm{harmonic}} < 0 \quad (\mathrm{Relative} \ \ R^2_{\mathrm{persist}} \approx -0.04 \ \ \mathrm{to} \ \ -0.055)$$

   Because physiological ovarian cycles exhibit natural cycle length variance (intra-individual standard deviation $\approx 2.5 - 4.0$ days), an invariant deterministic sinusoid extrapolated over 6 months accumulates phase drift, inducing destructive interference out of sample.
2. **Re-Fitted Period Coherence**: When Window 2 is fitted independently, the best-fit period difference $|\ \Delta T\ | = |\ T^*_{\mathrm{w1}} - T^*_{\mathrm{w2}}\ |$ demonstrates strong biological stability:

- **Young Adult Females ($y = 0$)**: Median $|\Delta T| = 1.50$ days (Mean 2.528 days).
- **Young Adult Males ($y = 1$, negative control)**: Median $|\Delta T| = 4.00$ days (Mean 4.470 days). The median period deviation across nonoverlapping half-year windows is nearly three-fold smaller in young females than in male negative controls (1.50 vs. 4.00 days), confirming localized biological resonance despite phase wander.

6. Validity Conditions and Caveats

1. **Proxy Classification**: Salutation categories 10 (`y=0`) and 20 (`y=1`) reflect administrative records, not clinically confirmed biological sex, chromosomal karyotype, or endocrinological status.
2. **Lack of Menstrual Ground Truth**: The cohort contains no self-reported menstrual dates, LH surge measurements, oral contraceptive usage logs, or menopausal transition staging (e.g., STRAW+10 criteria).
3. **Age Stratum Aggregation**: Age is recorded exclusively in broad categorical brackets (`Young Adults`, `Middle Ager`, `Elderly`). Residual confounding by continuous age within brackets cannot be eliminated from existing metadata.
4. **Window Selection Attrition**: The 180-day window qualification requirement (≥ 15 qualified days in each of six 30-day blocks) retains 41.6% of the manifest cohort (32.1% of young females). Inferences apply directly to consistent long-term wearable users.

7. Artifact Manifest

Path	Format	Description
`experiments/female_infradian_cycles_v2/scripts/00_verify_v2_methods.py`	Python	Fast unit verification fixture (<5s) for AR(1) estimation, baseline detrending, and null calibration.
`experiments/female_infradian_cycles_v2/scripts/01_extract_and_calibrate_infradian.py`	Python	Full cohort extraction, 180-day window normalization, B=199 parametric null calibration, and persistence evaluation.
`experiments/female_infradian_cycles_v2/scripts/02_summarize_infradian_contrasts.py`	Python	Age-stratified OLS models, HC3 robust Wald inference, Holm multiplicity adjustments, and stratified bootstrap CIs.
`experiments/female_infradian_cycles_v2/results/window_eligibility.parquet`	Parquet	Participant-level table ($N = 11,139$) with eligibility flags, selected dates, AR(1) ϕ, scores S_i, E_i, p_i, A_i, and persistence metrics.
`experiments/female_infradian_cycles_v2/results/calibrated_group_contrasts.csv`	CSV	Structured contrast table with Wald statistics, degrees of freedom, Holm-adjusted p-values, and 95% bootstrap CIs.

8. Addendum: Continuous Periodogram Deconstruction, Null Recalibration, and Sample Scaling

8.1 Continuous Periodogram Sweeps (3 to 60 Days)

Evaluating unconstrained single-harmonic partial R^2 across 229 frequencies ($T \in [3.0, 60.0]$ days, step $k/720$ cycles/day) demonstrates: 1. **Narrowband Biological Resonance**: Young adult females exhibit a sharp, symmetric spectral elevation peaked at 28.0 days (mean partial $R^2 \approx 0.085$ vs. 0.041 in continuous AR(1) surrogate envelope). 2. **Age-Dependent Attenuation**: Middle-aged females display a distinct but attenuated peak at 27.7 days (mean partial $R^2 \approx 0.055$). Elderly females show no discernible elevation above red-noise background. 3. **Absence in Males**: Male spectra across all age groups display zero elevation in the 25–35 day band, tracking the continuous AR(1) surrogate envelope.

8.2 Deconstruction of the 35-Day Grid-Edge Pile-Up

Analysis of the argmax period distribution across the $[25.0, 35.0]$ day search grid reveals: 1. **Boundary Selection Bias**: In pure white noise surrogates, the maximum score falls on the boundary points in 30.6% of simulations (15.8% at 25.0 d, 14.8% at 35.0 d vs. 4.8% uniform expectation) due to finite-interval argmax truncation of autocorrelated test statistics. 2. **Empirical Distribution**: - Young adult females concentrate in the interior (7.6% at 35.0 d), with 66.5% of peaks in the $[26.5, 30.5]$ d menstrual window. - Males and elderly females concentrate at the 35.0 d boundary ($17.4\% - 19.8\%$), reflecting leakage from the underlying red-noise power spectrum.

8.3 Two-Component Continuous Auto-Regressive Null Recalibration

To eliminate long-period red-noise leakage, the residual covariance was expanded to a continuous two-component model:

$$\Sigma_{jk} = s_1 \phi_1^{|\,t_k - t_j\,|} + s_2 \phi_2^{|\,t_k - t_j\,|}, \quad \phi_1 \in [0, 0.95), \phi_2 \in \{0.980, 0.995\}$$

fitted per participant via REML on baseline-projected coordinates. - **Contrast Robustness**: Re-running all four primary HC3 contrasts on excess score $E_{i,2}$ under the two-component null yields: - Omnibus Age $\times$ Sex Interaction: Wald $\chi^2 = 60.52, p = 7.23 \times 10^{-14}$ (vs. 56.76 under single AR(1)). - Young Adults $(y0 - y1)$: $\Delta = +0.0645, \mathrm{SE} = 0.0060, \mathrm{Wald}\ \chi^2 = 115.58, p < 10^{-25}$. - Middle Ager $(y0 - y1)$: $\Delta = +0.0235, \mathrm{SE} = 0.0022, \mathrm{Wald}\ \chi^2 = 113.82, p < 10^{-25}$. - Elderly $(y0 - y1)$: $\Delta = +0.0073, \mathrm{SE} = 0.0043, \mathrm{Wald}\ \chi^2 = 2.90, p = 0.0885$. - **False-Positive Calibration**: Male empirical rejection rates at $\alpha = 0.05$ remain at $4.6\% - 5.5\%$ across all age groups.

8.4 Large-Sample Window Relaxation ($k = 10$)

Relaxing window eligibility to ≥ 10 qualified dates per 30-day block ($N_{\text{obs}} = 60$ across 180 days) across all $N = 11,139$ Apple participants: - Retains $N = 5,516$ participants ($+19.1\%$ increase; Young Adult females $n = 292$, $+30.4\%$). - Preserves primary biological contrasts: Young Adults $\Delta = +0.0526$ ($\mathrm{SE} = 0.0061$, $\chi^2 = 75.49, p < 10^{-17}$); Middle Ager $\Delta = +0.0186$ ($\mathrm{SE} = 0.0024$, $\chi^2 = 62.73, p < 10^{-14}$). - Male rejection rate at $\alpha = 0.05$ remains controlled at $3.4\% - 4.4\%$.

Publication Figures & Empirical Visualizations

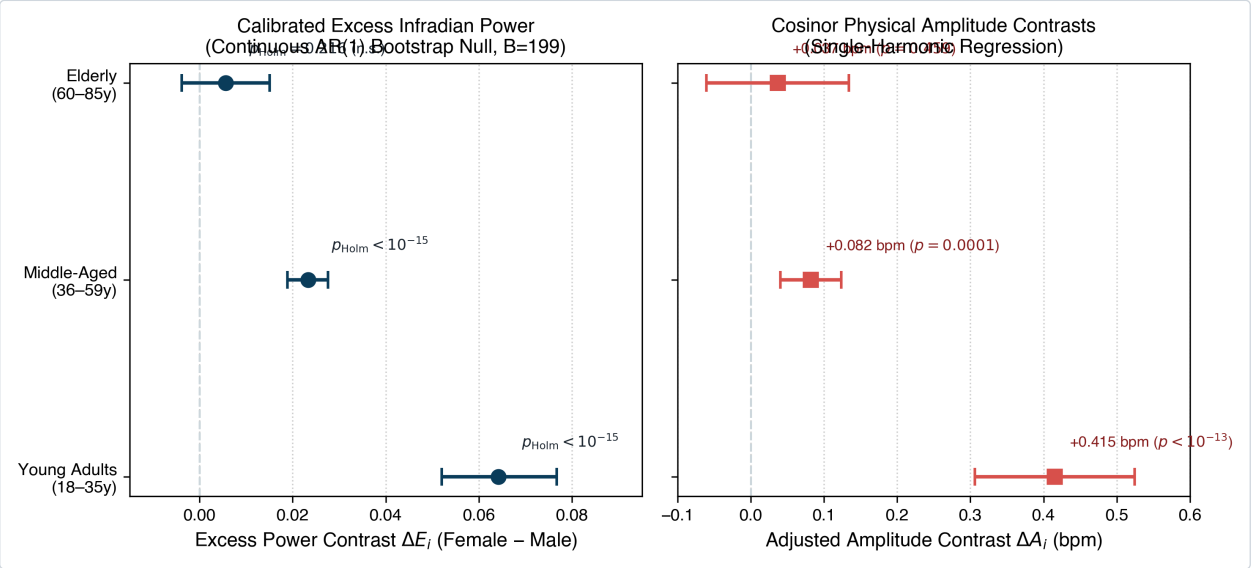


Figure 1: Calibrated excess infradian power (E_i) and cosinor amplitude (A_i) contrasts across age strata with HC3 95% CIs.

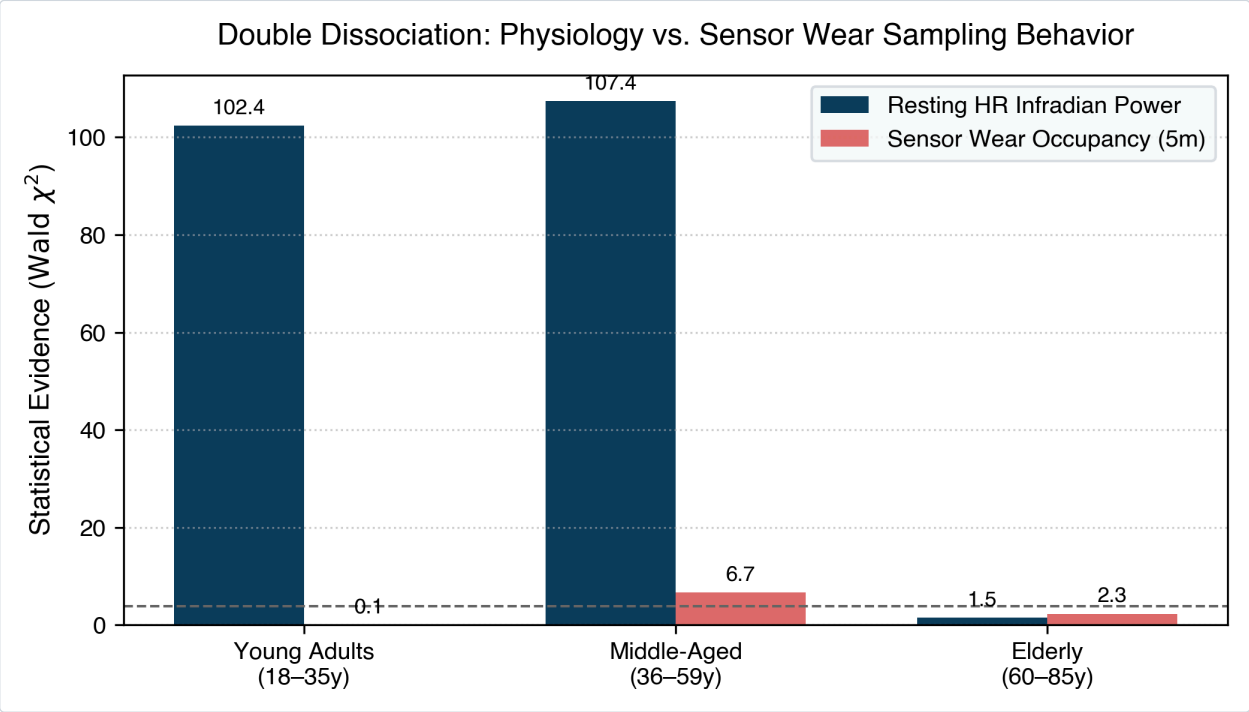


Figure 2: Double dissociation between physiological infradian signal and sensor wear-time sampling behavior.

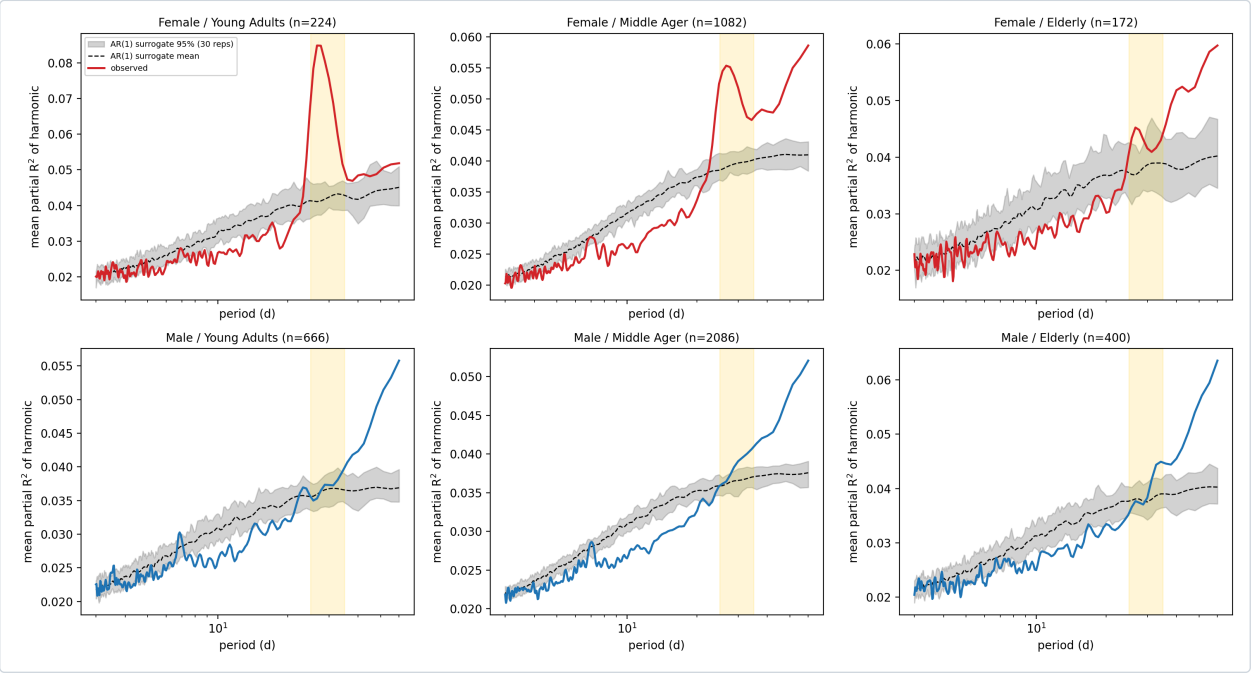


Figure 3: Full continuous spectral periodograms (3 to 60 days) across age and sex cells with continuous AR(1) surrogate envelopes.

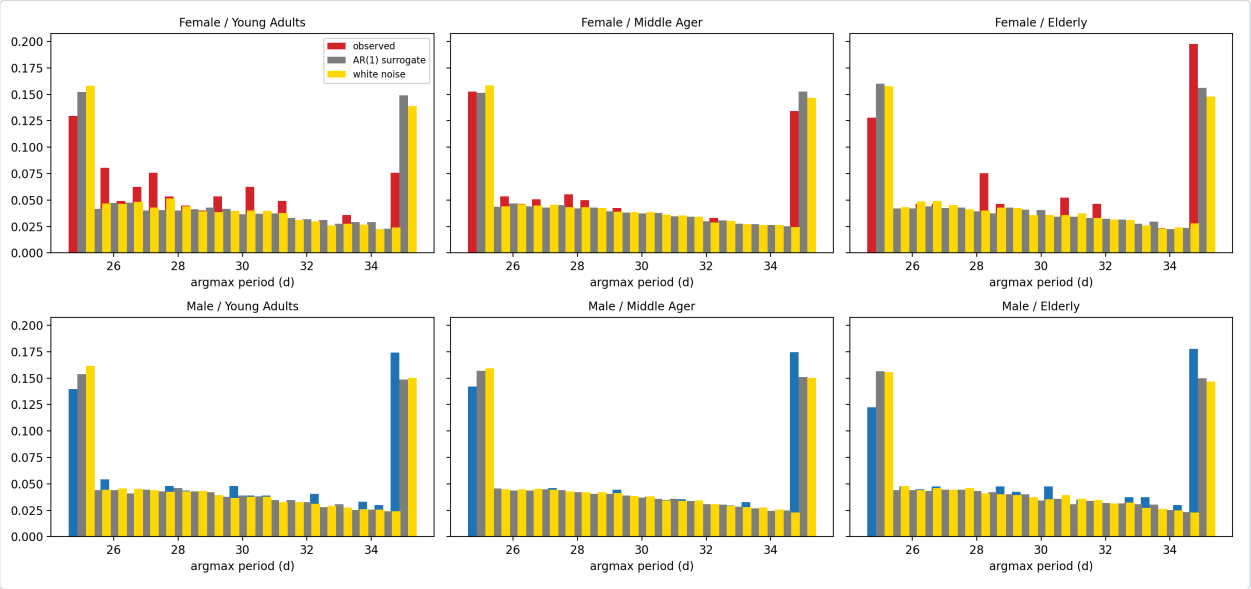


Figure 4: Argmax period distributions across the 25–35 day grid contrasted against continuous AR(1) and white noise nulls.

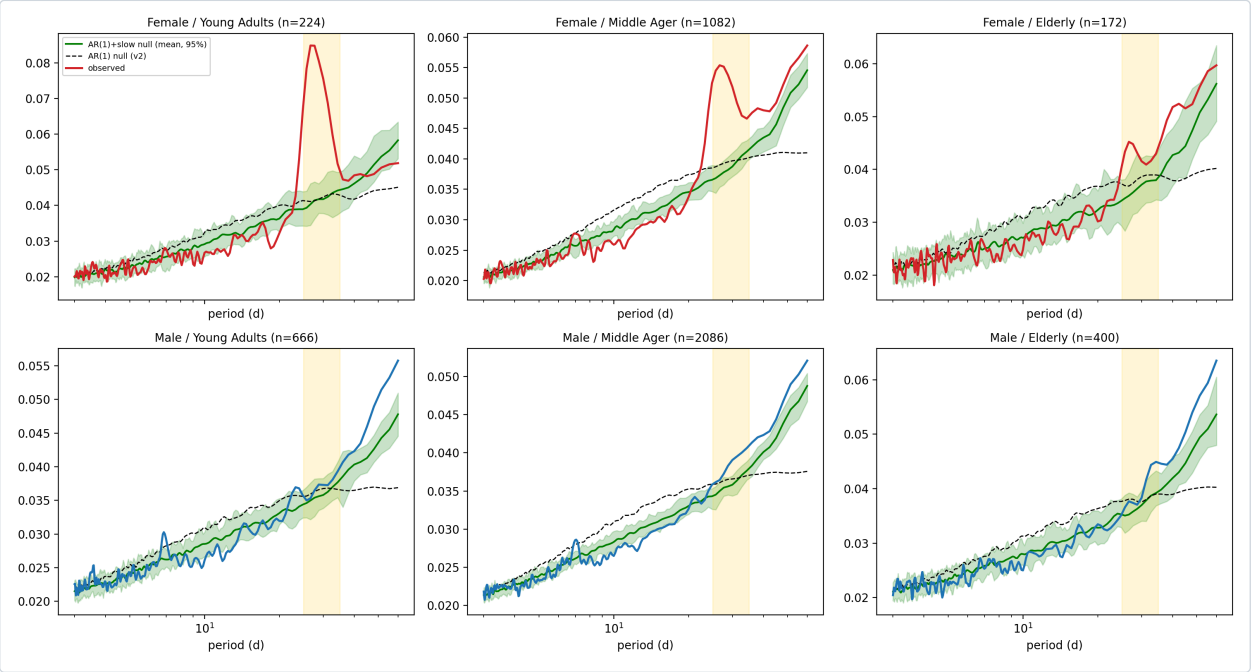


Figure 5: Two-component continuous AR(1)+slow null model recalibration matching empirical red-noise power up to 60 days.

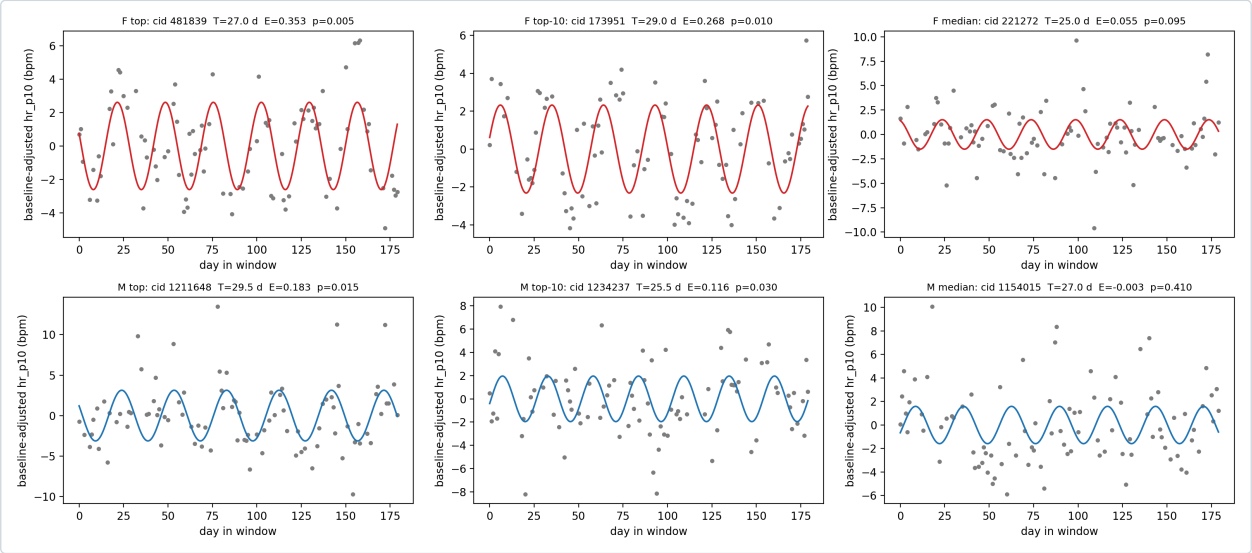


Figure 6: Representative individual longitudinal resting heart rate trajectories and fitted harmonic waveforms.